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A Catalog of 1.58 Million Clusters of Galaxies Identified from the DESI Legacy Imaging Surveys

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Abstract

Based on the DESI Legacy Imaging Surveys released data and available spectroscopic redshifts, we identify 1.58 million clusters of galaxies by searching for the overdensity of the stellar mass distribution of galaxies within redshift slices around preselected massive galaxies, with 877,806 clusters being found for the first time. The identified clusters have an equivalent mass of $M_{500} \geqslant 0.47 \times 10^{14}\, M_{\odot}$ with an uncertainty of 0.2 dex. The redshift distribution of clusters extends to $z \sim 1.5$, and 338,841 clusters have spectroscopic redshifts. Our cluster sample includes most of the rich optical clusters in previous catalogs, more than 95% of the massive Sunyaev–Zeldovich clusters, and 90% of the ROSAT and eROSITA X-ray clusters. From the light distributions of the member galaxies, we derive the dynamical state parameters for 27,685 rich clusters and find no significant evolution of the dynamical state with redshift. We find that the stellar mass of the brightest cluster galaxies grows by a factor of 2 since $z = 1$.

Unified Astronomy Thesaurus concepts: Galaxy clusters (584); Galaxy evolution (594); Catalogs (205)

Supporting material: machine-readable table

1. Introduction

As the largest virialized systems in the Universe, clusters of galaxies contain hundreds to thousands of galaxies, hot intracluster gas, and dark matter. They are tracers of large-scale structures and place powerful constraints on cosmological parameters (Allen et al. 2011; Böhringer et al. 2014; Hong et al. 2016). The dense environments and deep gravitational wells make galaxy clusters an ideal laboratory for understanding galaxy formation (Liu et al. 2008; Wetzel et al. 2012), emission of the intracluster medium (Rosati et al. 2002; Feretti et al. 2012), and the nature of dark matter (Markevitch et al. 2004; Banerjee et al. 2020).

Galaxy clusters can be identified as the overdensity peaks of the galaxy distribution in optical/infrared images (Abell 1958) or can be detected by the X-ray emission of the hot intracluster medium (Cavaliere & Fusco-Femiano 1976) or via the Sunyaev–Zeldovich (SZ; Sunyaev & Zeldovich 1972) signals in millimeter bands. The optical photometric surveys provide a large number of galaxy clusters. In the early years, about 10,000 nearby clusters were recognized from the single-band images of the Palomar Observatory Sky Survey (Zwicky et al. 1963; Abell et al. 1989; Gal et al. 2003; Lopes et al. 2004). Since multiband data of the Sloan Digital Sky Survey (SDSS) have been available, more than 150,000 galaxy clusters have been identified at redshifts $z < 0.8$ based on the red sequence of cluster member galaxies (Koester et al. 2007; Hao et al. 2010; Oguri 2014; Rykoff et al. 2014) or photometric redshifts of galaxies (Wen et al. 2009, 2012; Szabo et al. 2011; Wen & Han 2015; Banerjee et al. 2018). More clusters of galaxies have been found from the deeper data of the Hyper Suprime-Cam Subaru Strategic Program, the Dark Energy Survey (DES), and the Wide-field Infrared Survey Explorer (WISE) in the redshift range up to $z \sim 1.5$ (Oguri et al. 2018; Aguena et al. 2021; Wen & Han 2021, 2022; Thongkham et al. 2024).

Recently, the Dark Energy Spectroscopic Instrument (DESI) Legacy Surveys[4] have been carried out, and the released data in the three optical bands (g band, r band, and z band[5]) are one magnitude deeper than the SDSS (Dey et al. 2019). Incorporating the public DECam data, the DESI Legacy Surveys cover a sky area of more than 20,000 deg^2. Several large cluster catalogs have been published based on the previous DESI data release 9 (DR9; Yang et al. 2021; Zou et al. 2022; Yantovski-Barth et al. 2024). The latest DR10 data cover a larger sky area, with some overlapped areas with DR9 and including new i-band data from the NOIRLab Data Archive, an excellent database for identifying a larger sample of clusters to higher redshifts.

Following our previous work (Wen et al. 2009, 2012; Wen & Han 2015, 2021, 2022), in this paper we identify 1.58 million galaxy clusters from the released data of the DESI Legacy Surveys by using photometric and the available spectroscopic redshifts of galaxies. We first estimate photometric redshifts for galaxies with *griz* data from the DESI Legacy Surveys and mid-infrared data from WISE (Wright et al. 2010), as shown in Section 2. The details of identifying galaxy clusters and evaluating their physical parameters are given in Section 3. The comparisons with previously known clusters in the optical, X-ray, and SZ cluster catalogs are presented in Section 4. Cluster properties, such as dynamical states and the stellar mass of the brightest cluster galaxies

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[4] https://www.legacysurvey.org/
[5] We use z for redshift in this paper and z_m for the z-band magnitude, to avoid confusion.

(BCGs), are discussed in Section 5. A summary is given in Section 6.

Throughout this paper, we assume a flat Lambda cold dark matter cosmology, taking $H_0 = 70$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$. All magnitudes are given in the AB system.

2. Survey Data for Galaxies

The DESI Legacy Imaging Surveys (Dey et al. 2019) consist of three projects, including the Dark Energy Camera Legacy Survey (DECaLs), the Beijing–Arizona Sky Survey (BASS), and the Mayall z-band Legacy Survey (MzLS). The BASS and MzLS observe the northern hemisphere of decl. $\geqslant 32°$ with survey magnitude depths of $g = 24.0$, $r = 23.5$, and $z_{\rm m} = 22.9$, respectively, for point sources. The DECaLs covers the southern hemisphere of decl. $< 32°$ with magnitude depths of $g = 24.5$, $r = 23.9$, and $z_{\rm m} = 22.9$, respectively, deeper than the northern part (Zou et al. 2019). The southern survey part also includes data from the DES (Abbott et al. 2021) with magnitude depths of $g = 25.3$, $r = 25.0$, and $z_{\rm m} = 23.9$, respectively, data from the DECam Local Volume Exploration survey (Drlica-Wagner et al. 2022) with the depth of $z_{\rm m} = 22.8$; and data from the DECam eROSITA survey[6] with the depth of $z_{\rm m} = 23.2$. The DES survey data are deepest in the north Galactic cap region of $-9\overset{\circ}{.}5 \lesssim$ decl. $< 32°$, and are shallowest in the rest of the sky regions. The data of the DESI Legacy Surveys are supplemented with mid-infrared coadded imaging data from the unWISE (Lang 2014; Schlafly et al. 2019) with magnitude depths of W1 = 20.4 and W2 = 19.5, respectively.

The DESI Legacy Surveys DR9 contains the photometric data in the *grz* and W1W2 bands, covering a sky area of about 19,700 deg^2. The DR10 published data include objects from the DECaLs in the newly observed regions and also new *i*-band data for objects in some southern parts of DR9, covering a sky area of about 15,300 deg^2 with six bands (*griz*W1W2). The current DESI Legacy Surveys have released data for a total sky area of about 24,000 deg^2.

We analyze the DR9 and DR10 data independently for cluster identification in this work, because of the different number of bands, and choose the galaxy-type sources with TYPE $\neq$ PSF. The sources with bad photometry are discarded by using the following quality flags (Yang et al. 2021): FRAFLUX < 0.5, FRACMASKED < 0.4, and FRACIN > 0.3, in the *grz* bands for the DR9 data and the *griz* bands for the DR10 data. We also set a detection signal-to-noise ratio of S/N $\geqslant 5$ in the *grz*W1 and for the DR9 galaxies and the *griz*W1 bands for the DR10 galaxies. Some classified galaxies by their morphology are faint contaminated stars, which can be partly removed in the color–color space (Zhou et al. 2023) by setting $r - z_{\rm m} \geqslant 1$ and $z_{\rm m} - {\rm W1} < 1.2(r - z_{\rm m}) - 1.5$. Even so, some faint stars of $r - z_{\rm m} < 1$ cannot be distinguished from low-redshift galaxies. Fortunately, they will not affect our cluster identification work, because we use only massive galaxies and there are only a small number of such faint contaminated objects.

2.1. Spectroscopic Redshifts of Galaxies

The redshifts of galaxies are fundamental parameters for cluster identification in our algorithm (Wen & Han 2021, 2022). We adopt the spectroscopic redshifts of galaxies if they are available in the Two Micron All-Sky Survey (2MASS) Redshift Survey (Huchra et al. 2012), the SDSS DR17 (Abdurro'uf et al. 2022), and the early data release of the DESI (DESI Collaboration et al. 2023). These three redshift surveys give spectroscopic redshifts for 3.1 million galaxies in the DESI Legacy Surveys, which provide accurate redshifts for a large number of member galaxies for cluster identification.

2.2. Photometric Redshifts of Galaxies

The DESI Legacy Surveys DR9 includes photometric redshifts of galaxies estimated by the random forest algorithm using the *grz*W1W2 magnitude data (Zhou et al. 2021). We adopt these photometric redshifts for 361.1 million galaxies in the DESI Legacy Surveys DR9.

Following our previous papers (Wen & Han 2021, 2022), we estimate the photometric redshifts for galaxies in DR10 by using the nearest-neighbor algorithm, which is based on the empirical relation between galaxy colors and spectroscopic redshifts from a training sample. In the color spaces, the close neighbors of galaxies have a similar redshift. We calculate the distances in the color spaces between a targeted galaxy and all galaxies in the training sample. The photometric redshift of a targeted galaxy, $z_{\rm p}$, is estimated to be the median spectroscopic redshift of the 20 nearest neighbors. The uncertainty of the photometric redshift, $\sigma_{z_{\rm p}}$, is taken as the dispersion of these 20 spectroscopic redshifts.

For this work, we get a training sample of 883,000 galaxies with spectroscopic redshifts from the compiled data in Wen & Han (2021) and Wen & Han (2022) and also galaxies from the flux-limited data in the Galaxy And Mass Assembly (GAMA) Survey (Liske et al. 2015; Driver et al. 2022), the VIPERS DR2 (Scodeggio et al. 2018), and the DEEP2 (Matthews et al. 2013), plus the local galaxies in the 2MASS Redshift Survey (Huchra et al. 2012) and the galaxies in the early data release of the DESI (DESI Collaboration et al. 2023). The galaxies in this sample have a redshift in the range of $z < 1.6$.

We then estimate photometric redshifts for 295.6 million galaxies in the DESI Legacy Surveys DR10 with *griz*W1W2 magnitudes, see Table 1. In Figure 1, we compare the photometric redshifts with spectroscopic redshifts for galaxies with different z-band magnitudes in the region of middle survey depth. For bright galaxies of $z_{\rm m} < 21$, the uncertainty of the photometric redshifts, defined as $\sigma_{\Delta z} = 1.48 \times {\rm median}(|z_p - z_s|/(1 + z_s))$, is only about 0.013 for the galaxies at $z < 1$. The uncertainty increases to 0.028 for the galaxies of $21 < z_{\rm m} < 21.5$ at $0.7 < z < 1.3$ and to 0.038 for the galaxies of $21.5 < z_{\rm m} < 22$ at $0.9 < z < 1.5$. The photometric redshifts are reasonably estimated even for very faint galaxies of $z_{\rm m} > 22$, with an uncertainty of only 0.053 and a fairly small systematic bias of -0.041. The systematic bias and uncertainty are larger for fainter galaxies at $z > 1.2$, which are probably caused by the small number of faint galaxies at high redshifts in the training sample. Moreover, the 4000 Å break is moving out of the $z_{\rm m}$ band for objects of $z > 1.2$, making the estimate of photometric redshifts difficult. The survey depth does affect the accuracy of the photometric redshifts, especially for faint galaxies. As seen in Figure 2, the photometric redshift is slightly more accurate in the deeper region and worse in the shallower region for galaxies of $z_{\rm m} > 22$.

Based on the DESI Legacy Surveys DR9 data, Zhou et al. (2021) used the *grz*W1W2 magnitudes to estimate photometric redshifts. For galaxies of $z_{\rm m} < 21$, the redshifts are well estimated with an uncertainty of about 0.01332, but not for

[6] https://noirlab.edu/science/programs/ctio/instruments/Dark-Energy-Camera/DeROSITAS

Table 1
The Magnitudes, Estimated Photometric Redshifts, and Stellar Masses for Galaxies in the DESI Legacy Surveys DR10

R.A. (1)	Decl. (2)	g (3)	Err_g (4)	r (5)	Err_r (6)	i (7)	Err_i (8)	z_m (9)	Err_z (10)	W1 (11)	$\mathrm{Err}_{\mathrm{W1}}$ (12)	W2 (13)	$\mathrm{Err}_{\mathrm{W2}}$ (14)	z_p (15)	σ_{zp} (16)	z_s (17)	$\log(M_\star)$ (18)	Flag (19)
0.00002	−3.25165	23.427	0.082	22.989	0.069	22.733	0.104	22.274	0.105	21.287	0.182	22.241	0.755	1.2361	0.0774	−1.0000	10.01	0
0.00002	−0.27648	22.125	0.022	20.862	0.008	20.376	0.010	19.980	0.011	18.289	0.050	18.404	0.050	0.4243	0.0976	−1.0000	10.74	0
0.00002	−4.90665	23.219	0.049	22.688	0.039	22.158	0.041	21.682	0.040	20.421	0.081	20.590	0.205	1.1038	0.0791	−1.0000	10.33	0
0.00004	−1.09510	25.457	0.324	23.719	0.082	22.701	0.074	21.971	0.052	20.371	0.081	20.360	0.172	0.9334	0.0540	−1.0000	10.44	0
0.00004	−0.29959	22.046	0.021	21.152	0.012	20.969	0.019	20.709	0.024	19.690	0.051	20.880	0.284	0.4200	0.0695	−1.0000	9.87	0
0.00004	−0.62148	27.064	1.062	24.269	0.152	23.076	0.098	22.659	0.107	20.965	0.133	21.070	0.310	0.7433	0.0578	−1.0000	10.02	0
0.00007	−1.58000	24.180	0.100	23.386	0.058	22.645	0.067	22.463	0.078	21.247	0.170	23.265	1.362	0.7738	0.0250	−1.0000	9.55	0
0.00007	−4.43194	24.089	0.127	23.726	0.112	23.331	0.129	22.532	0.102	20.560	0.094	20.393	0.180	1.4287	0.0960	−1.0000	10.65	0
0.00007	−4.18086	23.170	0.062	22.282	0.033	21.400	0.023	21.063	0.028	19.550	0.050	20.062	0.136	0.8275	0.0357	−1.0000	10.55	0
0.00008	−2.42179	24.655	0.142	23.533	0.065	22.633	0.067	22.079	0.057	20.779	0.119	20.912	0.285	0.8729	0.0767	−1.0000	10.15	0

Note. The whole catalog is available at http://zmtt.bao.ac.cn/galaxy_clusters/. The columns are as follows. Columns (1) and (2): R.A. (J2000) and decl. (J2000) of a galaxy (in degrees). Columns (3)–(14): magnitudes and errors (AB system) in the *griz*W1W2 bands. Columns (15) and (16): photometric redshift and error. Column (17): spectroscopic redshift; −1.0000 means not available. Column (18): stellar mass in logarithm, with $M_\star$ in units of $M_\odot$. Column (19): flag for BCG-like galaxy; "1" for yes, "0" for no.

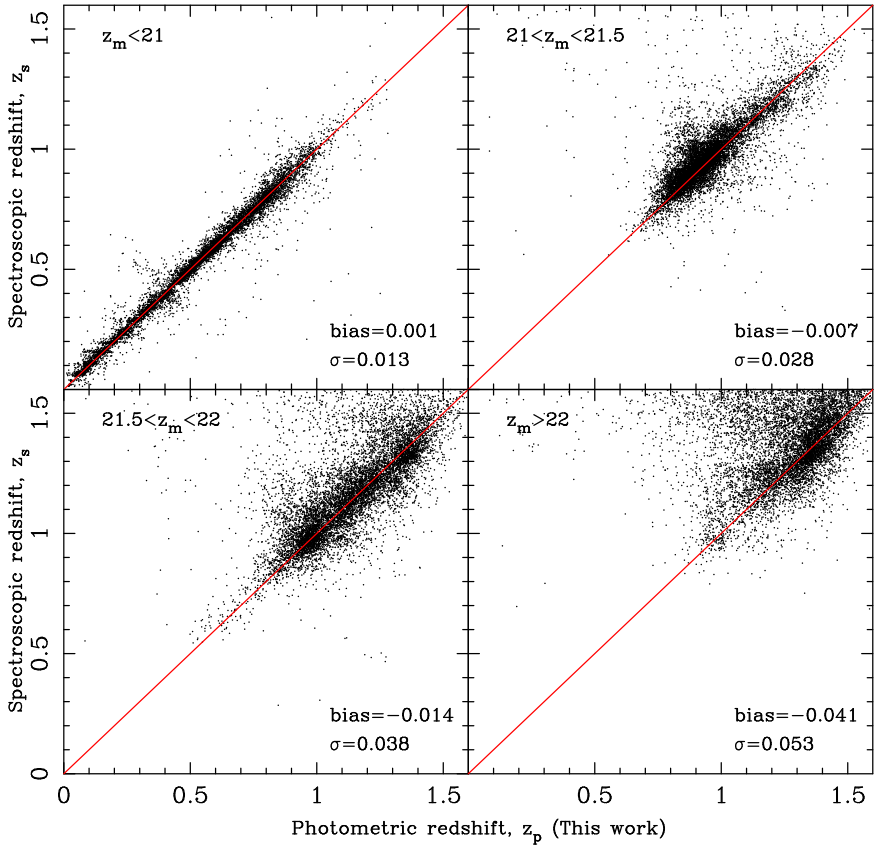


Figure 1. Comparison between the newly estimated photometric redshifts z_p and spectroscopic redshifts z_s for galaxies in the region of middle survey depth and in four z-band magnitude ranges, with different uncertainties and systematic bias, as marked in each panel, where only 10,000 randomly selected galaxies are taken for plotting.

fainter galaxies. For the galaxies of $z_m > 22$ in the shallow region of decl. $\geqslant 32°$, the photometric redshifts are systematically overestimated at $z_s < 1$ and underestimated at $z_s > 1$, with an uncertainty as large as 0.105 (see the right panel of Figure 2). We have tested our algorithm by ignoring the newly added i-band magnitude data for galaxies in the DR9 and DR10 and get consistent results with Zhou et al. (2021). We therefore conclude that the newly added i-band data in DR10 can significantly improve the accuracy of the photometric redshift estimates.

In summary, we take photometric redshifts for 361.1 million galaxies from the DESI Legacy Surveys DR9 and also the newly estimated photometric redshifts for 295.6 million galaxies from DR10 for cluster identification.

2.3. Stellar Masses of Galaxies

The stellar masses of galaxies constitute another fundamental parameter for cluster identification in our algorithm. The optical and infrared luminosities of galaxies are good tracers of their stellar mass (Bell et al. 2003; Kauffmann et al. 2003; Wen et al. 2013). In Wen & Han (2021), we took the values from the COSMOS2015 catalog (Laigle et al. 2016) to calibrate the relation between the W1-band luminosity, the color $r - z_m$, and the stellar mass of galaxies. Here, we do a similar calibration, but for the relation of the stellar mass with the z-band luminosity. We then obtain two kinds of stellar mass estimates for galaxies from the W1- and z-band luminosities separately and take the average. The final values are listed in Column (18) of Table 1. Compared to the stellar masses of galaxies in the COSMOS2015 catalog, our newly estimated masses have an uncertainty of only 0.15 dex (Figure 3), much improved from the single W1 band (Wen & Han 2021).

3. Galaxy Clusters Identified from the DESI Legacy Surveys

In the optical, galaxy clusters show a distinct overdensity of galaxies around a massive central galaxy. The total stellar mass of member galaxies is tightly related to the cluster mass (Andreon 2010; Pillepich et al. 2018; Palmese et al. 2020). Using the photometric redshifts, we can select most of the member galaxies with a small fraction of contamination (Wen et al. 2009). Therefore, we identify galaxy clusters by searching for an overdensity of stellar mass within a photometric redshift slice around massive galaxies.

3.1. Preselection for BCG-like Massive Galaxies

Galaxy clusters generally contain one or more massive galaxies, and the central one is called the BCG. Preselection for BCG-like galaxies can help to identify galaxy clusters more efficiently.

As the most massive galaxies in the Universe, the BCGs are very luminous in the optical and infrared bands (Hansen et al. 2009). In general, they contain an old stellar population and show a red color (Whiley et al. 2008), and are located at the bright end of the magnitude–redshift diagram and at the red end of the color–redshift diagram (Wen et al. 2018). We select

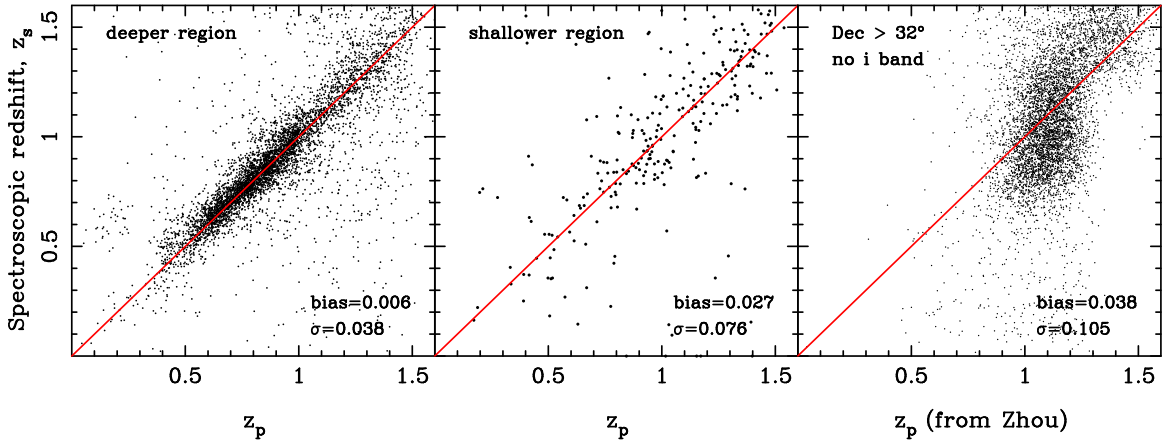


Figure 2. Comparison of photometric redshifts z_p with spectroscopic redshifts z_s for the faint galaxies of $z_m > 22$ in the deeper region (the left panel) and galaxies in the shallower region (the middle panel). The right panel is plotted for the galaxies in the shallower region of decl. $\geqslant 32°$ without *i*-band data, and the values of z_p are directly taken from Zhou et al. (2021). The uncertainty and systematic bias are marked in each panel.

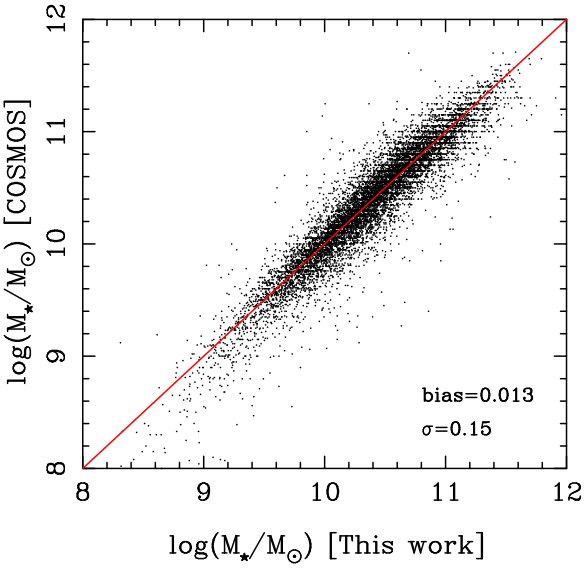


Figure 3. Comparison between our estimated stellar masses of galaxies with the values in the COSMOS2015 catalog.

BCG-like massive galaxies based on the *rz*W1 magnitudes and the estimated redshift and stellar mass.

First, the BCG-like galaxies are selected to have a stellar mass of $M_\star \geqslant 10^{11}\,M_\odot$ (Lidman et al. 2012; Wen & Han 2021). Then, according to the distributions of known BCGs from the Abell clusters (Lauer et al. 2014) and the WHL and WH22 clusters (Wen et al. 2012; Wen & Han 2015, 2022), we set the criteria shown in Figure 4, which cause a loss of 2.5% of the known faint BCGs but exclude many fainter galaxies detected in the *z* and W1 bands. Similarly, some blue massive galaxies are also discarded according to the threshold shown in the color–redshift diagram (r–z_m) in Figure 4. Note that BCGs at low redshifts have a tight distribution in colors, but these at high redshifts have much more scattered colors due to star formation (O'Dea et al. 2008; Liu et al. 2012; McDonald et al. 2016; Wen & Han 2021).

In total, 15.23 million galaxies are preselected as possible BCGs, as indicated in Column (19) of Table 1, of which 9.40 million galaxies have *i*-band data.

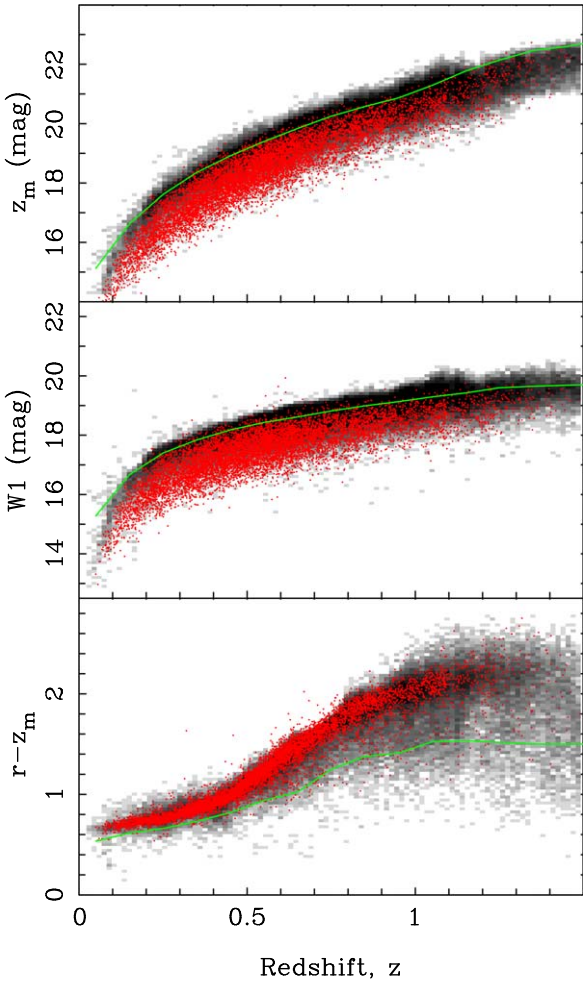


Figure 4. Magnitudes at the *z* and W1 bands and color ($r - z_m$; bottom) against redshift for 10,000 randomly selected known BCGs, as indicated by the small (red) dots. The distribution of all massive galaxies of $M_\star \geqslant 10^{11}\,M_\odot$ in the DESI Legacy Surveys is shown with the number density in gray. We set criteria for the BCG-like galaxies as shown by the solid line.

3.2. Identification of Galaxy Clusters

Following our previous works (Wen & Han 2021, 2022), we identify galaxy clusters by searching for an overdensity of

stellar mass around massive galaxies within a redshift slice and then estimating the parameters of the cluster candidates.

Previously, to get the cluster radius and mass for galaxy clusters to be identified, we first calibrated the scaling relation between the total stellar mass within a given radius and the cluster radius (r_{500}) or mass (M_{500}) by using a sample of M_{500} known clusters. Here, r_{500} is the radius within which the mean density is 500 times the critical density of the universe. M_{500} is the cluster mass within r_{500}. We took the galaxies within a photometric redshift slice, $z \pm \Delta z$, around massive galaxies as member galaxy candidates for cluster candidates. Half of the slice thickness is taken as being

$$\Delta z = \begin{cases} 0.04\,(1+z) & \text{for } z \leqslant 0.7 \\ 0.15\,z - 0.037 & \text{for } z > 0.7 \end{cases}. \quad (1)$$

We discarded the member galaxy candidates with a large redshift uncertainty of $\sigma_{zp} > 2\,\Delta z$. Then, we searched for galaxy cluster candidates from all the photometric data. The redshift of the cluster candidate was estimated as the median value of the photometric redshifts of the member galaxy candidates. The cluster radius and richness (a mass proxy) were estimated according to the total stellar mass of the member galaxy candidates and the calibrated scaling relations. A galaxy cluster was identified if it is above a richness threshold.

In this work, we make some improvements to our procedures. First, only massive galaxies with $M_\star \geqslant 10^{10}\,M_\odot$ are used, since they have a smaller redshift uncertainty than these less massive (and fainter) galaxies. The central BCGs of the clusters to be identified are therefore only taken from the preselected BCG-like galaxies. Second, the spectroscopic redshifts of some galaxies, if available, are used to determine the redshifts of the cluster candidates and to label member galaxies. We adopt the available spectroscopic redshift of the BCG as the spectroscopic redshift of a cluster. If the spectroscopic redshift of the BCG is not available, the available spectroscopic redshifts of other galaxies are adopted if they are within $0.025(1+z)$ of the cluster photometric redshifts. The member galaxies with spectroscopic redshifts are determined if they have a velocity difference $\Delta v < 2500$ km s^{-1} from the cluster spectroscopic redshifts. Third, we verify the scaling relations of r_{500} and M_{500} by using a large cluster sample (Wen & Han 2015), in which the cluster masses have been rescaled for consistency and calibrated to weak lensing measurements (Vikhlinin et al. 2009). As done in our previous work (Wen & Han 2021), we get the scaling relation for r_{500} as

$$\log r_{500} = 0.402 \log \frac{M_{\star,r1}}{10^{10} M_\odot} - (0.944 \pm 0.03) + (0.43 \pm 0.02)\log(1+z), \quad (2)$$

where $M_{\star,r1}$ is the total stellar mass (after background subtraction) of the member galaxy candidates within a projected radius of $r_1 = 1.0\,E(z)^{-2/3}$Mpc, where $\mathrm{E(z)^2}=\Omega_\Lambda+\Omega_\mathrm{m}(1+z)^3$. The total stellar mass, $M_{\star,500}$, is then calculated from galaxies within a projected radius of r_{500}. We get the scaling relation between $M_{\star,500}$ and the cluster mass M_{500}, which is consistent with our previous result (Wen & Han 2021). The richness of a galaxy cluster, λ_{500}, is then defined as a redshift-independent mass proxy by

$$\lambda_{500} = M_{\star,500}(1+z)^{0.21}/M_\star^*, \quad (3)$$

where $M_\star^*$ is a characteristic stellar mass. A galaxy cluster is identified when it has a richness of $\lambda_{500} \geqslant 10$ and the number of member galaxy candidates within r_{500} is $N_{\rm gal} \geqslant 6$. The threshold is slightly lower than that we adopted in Wen & Han (2021), so we do find more low-mass clusters.

After cleaning the repeated entries of identified clusters, we finally get 1,581,179 galaxy clusters, as listed in Table 2. Among them, 877,806 clusters are identified for the first time; 946,486 clusters are identified by using our newly obtained photometric redshifts, while the other 634,693 clusters are identified from the photometric redshifts in the DESI Legacy Surveys DR9 data. The 15,864,358 member galaxy candidates are publicly available at http://zmtt.bao.ac.cn/galaxy_clusters/.

The redshift distribution of our clusters extends to $z \sim 1.5$, and is compared to those clusters previously found from the DESI Legacy Surveys DR9 (Yang et al. 2021; Zou et al. 2022; Yantovski-Barth et al. 2024): see Figure 5.

The sky distribution of galaxy clusters is shown in Figure 6. The clusters in the DES region in the south Galactic cap have the highest number density, being ∼108 clusters per deg^2. The density declines to ∼80 clusters per deg^2 in the north Galactic cap region with $-9.^\circ5 \lesssim$ decl. $< 32^\circ$ and to about ∼50 clusters per deg^2 in the rest of the regions.

3.3. Verification of Clusters and Their Parameters

To validate the reliability of the identified clusters, we crossmatch our clusters in the deep COSMOS field with previously known clusters. The COSMOS has a field of view of 2 deg^2 and more than 30 bands for deep imaging observations at wavelengths from the NUV (e.g., Zamojski et al. 2007) and optical (e.g., Scoville et al. 2007; Taniguchi et al. 2007) to infrared (e.g., Sanders et al. 2007). Deep X-ray observations were also carried out for this sky area by the Chandra and XMM-Newton (e.g., Hasinger et al. 2007; Civano et al. 2016). Based on these high-quality data, galaxy clusters/groups have been identified with high completeness at low redshifts (Wen & Han 2011, 2015; Knobel et al. 2012; Söchting et al. 2012; Darvish et al. 2017; Gozaliasl et al. 2019). Most of the clusters in the field have spectroscopic observations.

In the COSMOS field, we detect 221 clusters, 192 of which have counterparts in the previous catalogs. At $z < 0.9$, only four of 145 clusters are new. We carefully inspect all 221 clusters in the DESI color images and find that they all have a concentration of galaxies around their BCGs with a similar color. We conclude that our cluster sample has a high purity.

In our cluster sample, 338,841 clusters have spectroscopic redshifts for one or more galaxies. To test the accuracy of the cluster photometric redshifts, we perform our algorithm by using the photometric redshifts only for these clusters. Figure 7 shows the comparison between the cluster photometric redshifts and spectroscopic redshifts. One can get the uncertainty of the photometric redshift, $\sigma_z^{\rm cl} = 1.48 \times$ median $(|z_{\rm cl} - z_{\rm s}|/(1+z_{\rm s})) = 0.010$, for clusters identified from the data with available i-band magnitudes (see the left panel of Figure 7), while it is $\sigma_z^{\rm cl} = 0.012$ for the clusters identified from the data without i-band magnitudes in the shallowest region of decl. $\geqslant 32^\circ$ (see the right panel of Figure 7).

Most member galaxy candidates are obtained by photometric redshifts, except for those with spectroscopic redshifts. The accuracy and completeness of the detected member galaxies are tested by using the spectroscopic survey data of ∼300,000

Table 2
A Catalog of 1.58 Million Clusters of Galaxies Identified from the DESI Legacy Surveys

ID (1)	Name (2)	R.A. (3)	Decl. (4)	z_{cl} (5)	$flag_z$ (6)	$z_{m,BCG}$ (7)	$W1_{BCG}$ (8)	$\log(M_{\star,BCG})$ (9)	r_{500} (10)	λ_{500} (11)	M_{500} (12)	N_{gal} (13)	Γ (14)	σ_Γ (15)	Source (16)	Other Catalog (17)
1	J000000.1-372835	0.00046	−37.47628	0.1364	0	15.377	15.561	11.49	0.766	25.79	1.17	14	9.99	9.99	1	WHY18
2	WH-J000000.3-563518	0.00135	−56.58835	0.1957	0	15.712	15.723	11.70	0.625	23.24	1.06	7	9.99	9.99	1	…
3	J000000.4-391925	0.00172	−39.32369	0.3580	0	17.713	17.313	11.41	0.526	15.90	0.73	7	9.99	9.99	1	Y21
4	J000000.5+021911	0.00201	2.31980	0.4282	1	18.043	17.359	11.55	0.612	23.40	1.07	13	9.99	9.99	1	redMaPPer
5	WH-J000000.6-590047	0.00260	−59.01297	0.1380	0	14.993	15.186	11.69	0.606	15.71	0.73	7	9.99	9.99	1	…
6	J000000.8-412726	0.00327	−41.45719	0.4015	0	18.386	17.691	11.29	0.462	10.95	0.51	6	9.99	9.99	1	CFSFDP
7	J000001.0+012416	0.00403	1.40456	0.6538	1	18.611	17.088	11.85	0.560	24.77	1.13	6	9.99	9.99	1	CFSFDP
8	WH-J000001.0-594933	0.00426	−59.82570	0.6250	0	19.248	18.155	11.29	0.457	15.18	0.70	8	9.99	9.99	1	…
9	J000001.0-535824	0.00435	−53.97320	0.6112	0	18.693	17.541	11.66	0.607	23.42	1.07	11	9.99	9.99	1	redMaPPer
10	J000001.2-030053	0.00504	−3.01485	0.5090	1	18.241	17.339	11.67	0.543	21.71	0.99	8	9.99	9.99	1	Y21

Notes. Column (1): cluster ID. Column (2): cluster name with J2000 coordinates—the names with "WH" are newly identified in this paper. Columns (3) and (4): R.A. (J2000) and decl. (J2000) of the cluster BCG (in degrees). Column (5): cluster redshift z_{cl}. Column (6): redshift flag; "0" for photometric redshift in Column (5) and "1" for spectroscopic redshift. Columns (7)–(8): BCG magnitudes (AB system) in the z and W1 bands, respectively. Column (9): logarithm of BCG stellar mass, with $M_{\star,BCG}$ in units of $M_\odot$. Column (10). Cluster radius, r_{500}, in megaparsecs. Column (11): cluster richness. Column (12): derived cluster mass, in units of $10^{14}\,M_\odot$. Column (13): number of member galaxy candidates within r_{500}. Columns (14)–(15): relaxation parameter and error for rich clusters with $N_{gal} \geqslant 30$, otherwise 9.99 is given. Column (16): data source for the cluster; "1" for the data with i-band magnitudes available and "2" for the data without i-band magnitudes. Column (17): reference notes for previously known clusters: maxBCG (Koester et al. 2007); WHL (Wen et al. 2009, 2012; Wen & Han 2015); GMBCG (Hao et al. 2010); AMF (Szabo et al. 2011; Banerjee et al. 2018); redMaPPer (Rykoff et al. 2014, 2016); CAMIRA (Oguri 2014); WHY18 (Wen et al. 2018); WH21 (Wen & Han 2021); Y21 (Yang et al. 2021); WaZP (Aguena et al. 2021); WH22 (Wen & Han 2022); CFSFDP (Zou et al. 2021, 2022); and CluMPR (Yantovski-Barth et al. 2024).

(This table is available in its entirety in machine-readable form.)

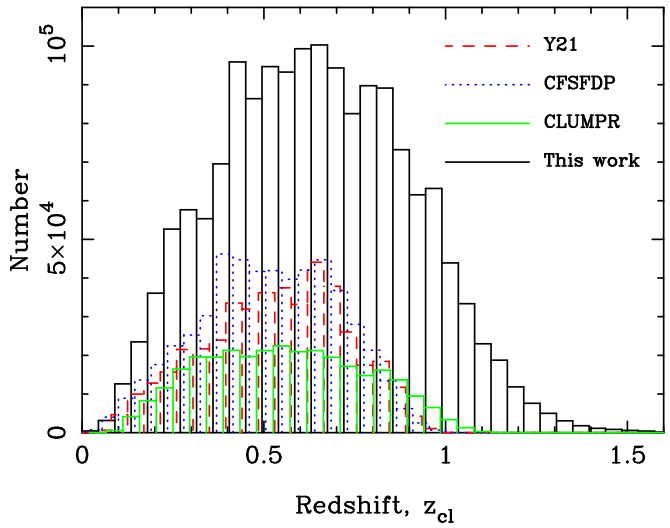


Figure 5. Redshift distribution of the identified 1,581,179 clusters, compared to clusters previously identified from the DESI Legacy Surveys DR9, including Y21 (Yang et al. 2021), CFSFDP (Zou et al. 2022), and CLUMPR (Yantovski-Barth et al. 2024).

galaxies down to flux-limited $r < 19.8$ mag from the GAMA survey covering $\sim$286 deg^2 (Driver et al. 2022). Those galaxies with a velocity difference of less than 2500 km s^{-1} from the BCGs are taken as true member galaxies. The member galaxy candidates of clusters determined by photometric redshifts only are crossmatched with the spectroscopic data of the GAMA survey. We find that 78% of the member galaxy candidates can be verified by spectroscopic redshifts if the BCGs are not included or up to 83% if the BCGs are included. As seen in Figure 8, more massive galaxies have a higher fraction of true members detected. Due to the uncertainty of the photometric redshifts, some true members are not included in the redshift slices. The missing member galaxies are such galaxies with a velocity difference of less than 2500 km s^{-1} from the BCGs. The missing fraction is about 8% of the true members, slightly higher for lower-mass galaxies (see Figure 8). Note that the GAMA survey is much shallower compared to the DESI Legacy Surveys, so this test is made only for the bright galaxies in the DESI Legacy Surveys.

The cluster richness λ_{500} is compared to the mass M_{500} for the common clusters in Wen & Han (2015). As shown in Figure 9, they are well correlated and can be scaled by

$$\log M_{500} = (0.97 \pm 0.03)\log \lambda_{500} - (1.30 \pm 0.02), \quad (4)$$

where M_{500} is in units of $10^{14}\,M_\odot$. The richness threshold of $\lambda_{500} = 10$ corresponds to the mass of $M_{500} \sim 0.47 \times 10^{14}\,M_\odot$. By Equation (4), M_{500} can be estimated by λ_{500} with an uncertainty of 0.20 dex for all identified clusters.

4. Crossmatching Clusters with Those in Previous Catalogs

In the sky area of the DESI Legacy Surveys, some survey data have previously been released, from which a large number of clusters of galaxies have been identified in the literature, as marked in the last column of Table 2.

In the following, we crossmatch the identified clusters with those in some defined redshift ranges and sky regions in previous cluster catalogs, including the optical clusters from the DESI Legacy Surveys and DES, the X-ray clusters from the ROSAT All-Sky Survey (RASS) and the extended ROentgen Survey with an Imaging Telescope Array (eROSITA), and the SZ clusters from Planck, the South Pole Telescope (SPT), and the Atacama Cosmology Telescope (ACT).

4.1. DES and DESI Legacy Surveys Clusters

Rykoff et al. (2014) identified galaxy clusters based on the red-sequence feature of cluster galaxies and presented the redMaPPer cluster catalog. The red-sequence model was calibrated by using a training sample and then applied to photometric data to search for clusters. From the DES Y1 data covering a sky area of $\sim$1800 deg^2, 6729 redMaPPer clusters in the redshift range of $0.2 < z \lesssim 0.8$ were identified (Rykoff et al. 2016). Crossmatching with our clusters in the same sky region and redshift range shows that 95% of redMaPPer clusters can be detected within a redshift difference of $0.05(1+z)$ and a projected distance of $1.5\,r_{500}$ from our clusters. As shown in Figure 10, the detection fraction depends on the richness, and nearly 100% of clusters with a richness of $\lambda > 50$ are detected. Understandably, some clusters with small richness are missed due to the uncertainties of the photometric redshifts of galaxies and hence the member recognition.

Wen & Han (2022; WH22) applied the same algorithm as in this paper to the DES data and identified 151,244 clusters at $0.1 < z < 1.5$. About 94% of the clusters at $z < 1$ can be found in the catalog of this work. The detection fraction is near 100% for rich clusters of $\lambda > 50$ (see Figure 10).

Three cluster samples obtained from photometric redshifts of the DESI Legacy Surveys DR9 have also been cross-matched with our catalog. Yang et al. (2021; Y21) used the friends-of-friends algorithm and identified 5.8 million groups in the redshift range of $z < 1$ that have at least three members.[7] Zou et al. (2022) applied "the Clustering by Fast Search and Find of Density Peaks (CFSFDP)" algorithm and identified 532,810 clusters (Zou 2023) also in the redshift range of $z < 1$. Yantovski-Barth et al. (2024) presented "the Clusters from Masses and Photometric Redshifts (CluMPR)" algorithm and found 309,115 clusters (Yantovski-Barth et al. 2023) in the redshift range of $0.1 < z < 1$. The Y21 catalog contains many galaxy groups down to a very low mass. We here choose the 444,149 clusters with halo masses (M_h) above our threshold after the mass conversion function $M_{500} = 0.55\,M_h$ (Hu & Kravtsov 2003) is applied. In our cluster catalog, we detect 92% of the Y21 clusters, 72% of the CFSFDP clusters and 86% of the CluMPR clusters, respectively. The detection fractions vary as a function of cluster mass or richness.

Notice that some clusters in previous catalogs are not detected by our algorithm. We investigate and find three reasons. The first is that a cluster is located in a cluster–binary system, probably just starting the merging process. We find both, but their small separation ($<1.5\,r_{500}$) leads us to regard them as one cluster. The poorer one is removed during our cleaning stage but is listed in the other catalogs with a different cluster center. Second, the unmatched clusters in previous catalogs have no BCG-like galaxies. Third, the rest are clusters with BCG-like galaxies, but their richness and the total number of member galaxy candidates are below our thresholds.

[7] https://gax.sjtu.edu.cn/data/DESI.html

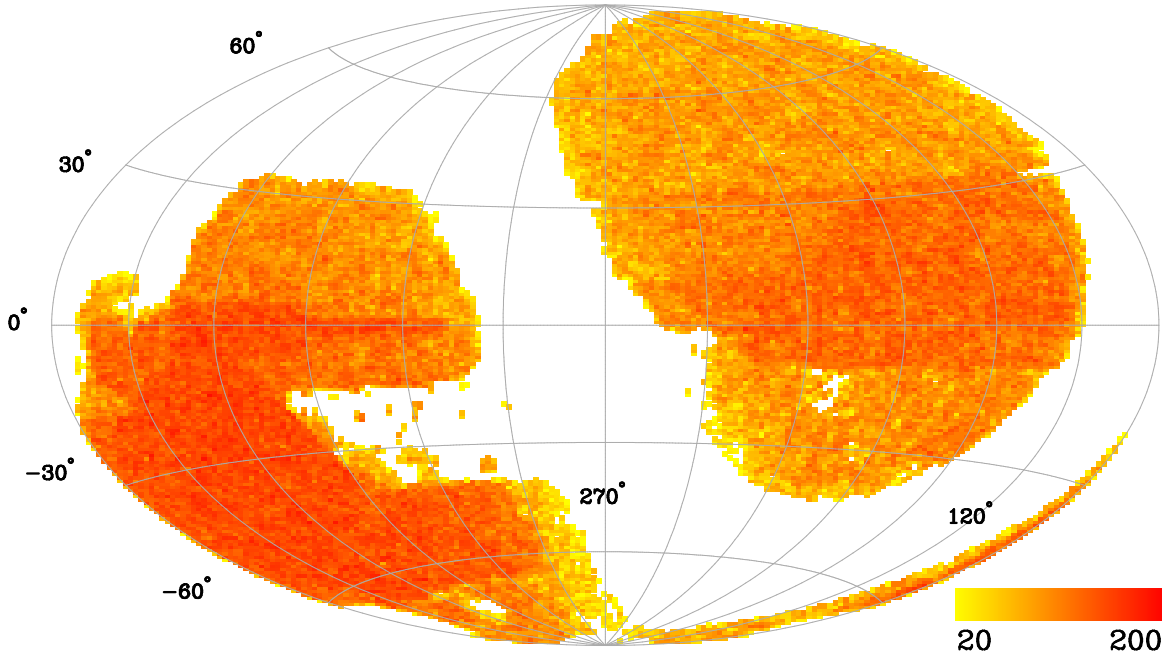


Figure 6. The sky distribution of the identified galaxy clusters in the Equatorial coordinate system, expressed by the number of clusters per square degree.

4.2. X-Ray Clusters

The X-ray emission of galaxy clusters comes from hot intracluster gas. Both the hot gas and galaxy components trace the mass distribution inside a cluster. In principle, any optical cluster should have a counterpart in X-ray. We crossmatch our clusters from the DESI Legacy Surveys with the X-ray clusters identified from RASS (Klein et al. 2023) and eROSITA (Bulbul et al. 2022; Liu et al. 2022). We find that most ROSAT clusters are massive, while the eROSITA clusters are much less massive, as shown in Figure 11.

Klein et al. (2023) presented a large X-ray cluster sample based on a joint analysis of the second RASS source catalog (2RXS) and the DESI Legacy Surveys data. They searched for optical counterparts of the 2RXS sources by using a multi-component matched filter (MCMF) algorithm (Klein et al. 2018). The redshift, optical richness, and X-ray luminosity are derived, and a probability of being a random superposition $f_{\rm cont}$ is given. The RASS-MCMF catalog contains 8449 X-ray clusters at $z < 1$ with a purity of 90%. In the sky coverage of this work, there are 7898 RASS-MCMF clusters. We find that 90% of them are detected in this paper. The matched and unmatched RASS-MCMF clusters are shown in the mass–redshift diagram in Figure 11. The detection fractions increase to 93% and 95% for the RASS-MCMF subsamples with purities of 95% and 99%, respectively.

The eROSITA all-sky survey is 25 times more sensitive than RASS, and it aims to detect about 50,000–100,000 galaxy clusters (Merloni et al. 2012). The data release of the eROSITA Final Equatorial Depth Survey (eFEDS) covers an area of $\sim$140 deg^2. Liu et al. (2022) detected 542 candidates of galaxy clusters and groups at $0.01 < z < 1.3$ with an extended X-ray morphology. The sample has a completeness of 40% and a purity of 80% down to the flux limit of 10^{-14} erg s^{-1} cm^{-2}. We take a subsample of 262 clusters and groups with an extent likelihood $\geqslant$15 that has a purity of 90%. We find that 91% of the eROSITA clusters and groups are detected in this work. The X-ray candidates of galaxy clusters and groups with a smaller extent likelihood may be easily misclassified from point sources. Another eFEDS cluster catalog contains 346 galaxy clusters and groups in the redshift range of $0.1 < z < 1.3$ with a mass of $>10^{13}\,M_{\odot}$ (Bulbul et al. 2022), of which 92% clusters and groups can be found in our sample (see Figure 11). After our paper was submitted, the first eROSITA All-Sky Survey (eRASS1) published the largest X-ray cluster catalog, which contains 12,247 clusters in the western Galactic hemisphere (Bulbul et al. 2024), with a purity of 86%. About 82% of eRASS1 clusters are detected in our sample. The crossmatching indicates that our optical sample has a good completeness, even for low-mass clusters of about $5 \times 10^{13}\,M_{\odot}$.

We check the data and also visually inspect the color images of the DESI Legacy Surveys to investigate why some X-ray clusters are missing in our cluster catalog. We find that some of them are merging clusters, with a projected distance between the optical and X-ray positions larger than the matching radius. Otherwise, the BCGs in some clusters are not found due to the cutoff caused by photometry flags. For most of the low-mass clusters, we can see the overdensity of the member galaxy candidates, but the BCG stellar mass or the richness estimated from the member galaxy candidates is lower than our thresholds.

4.3. SZ Clusters

The SZ effect is the result of inverse Compton scattering of the cosmic microwave background by the hot intracluster gas. The surface brightness of the SZ effect is independent of redshift, so the SZ effect has the advantage of showing clusters at high redshifts (Carlstrom et al. 2002).

The PLANCK satellite carried out an all-sky millimeter survey in the millimeter bands (Planck Collaboration et al. 2014). The second release of the PLANCK SZ catalog contains 1653 massive clusters with $M_{500} > 2 \times 10^{14}\,M_{\odot}$ (Planck Collaboration et al. 2016), of which 789 clusters are located in the DESI Legacy Surveys footprint. We find 746 matched

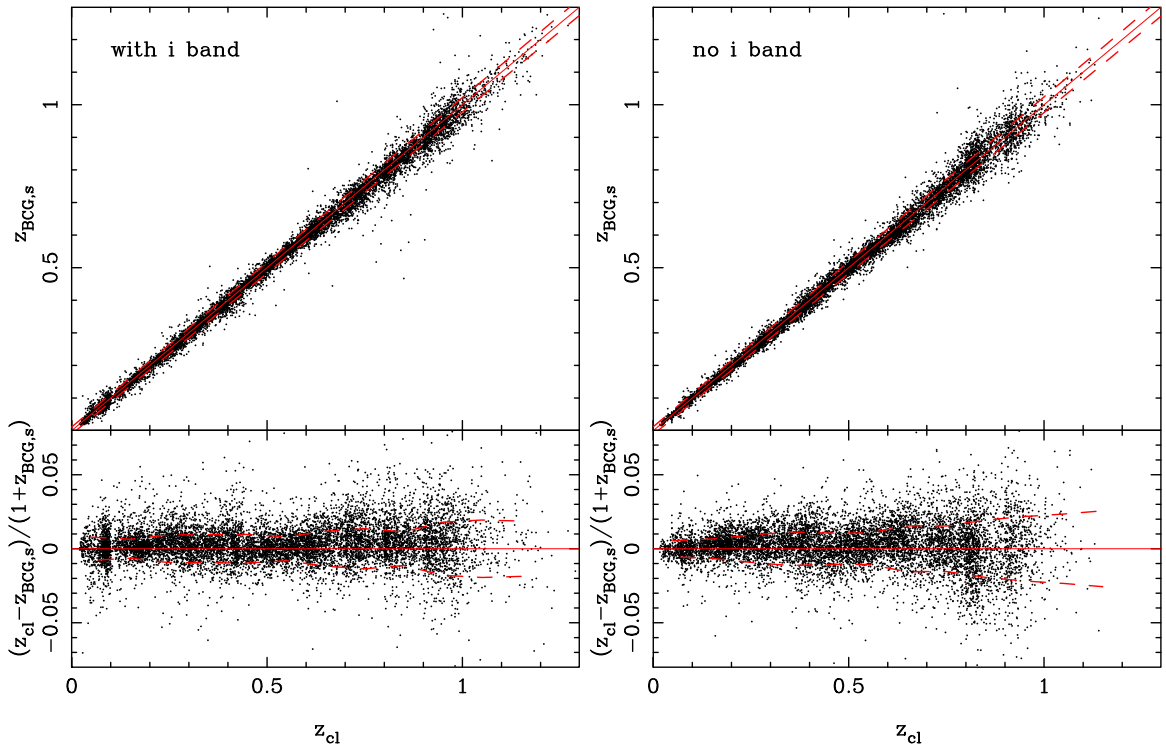


Figure 7. The left panel shows the comparison between photometric redshifts and spectroscopic redshifts for clusters identified with *i*-band data. In every redshift interval of 0.1, 1000 clusters are randomly selected. The solid line is for equal values, while the dashed lines represent the deviation of $\pm 1\sigma$. The right panel is the same as the left, but for clusters identified without *i*-band data in the shallowest region of decl. $\geqslant 32°$.

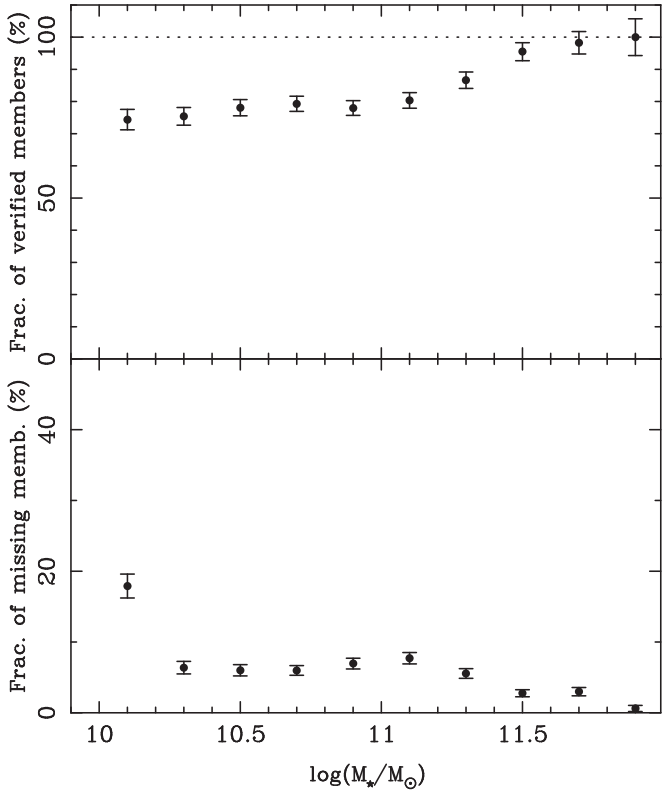


Figure 8. Verification of cluster members as a function of the stellar mass of galaxies by using the spectroscopic survey data of GAMA (Driver et al. 2022). Upper panel: the fraction of verified member galaxies ($N_{veri.}/N_{all}$). Lower panel: the fraction of missed true member galaxies ($N_{missing}/N_{real}$). Here, $N_{veri.}$, $N_{missing}$, and N_{real} are the numbers of galaxies defined within a small spectroscopic redshift offset from the BCGs.

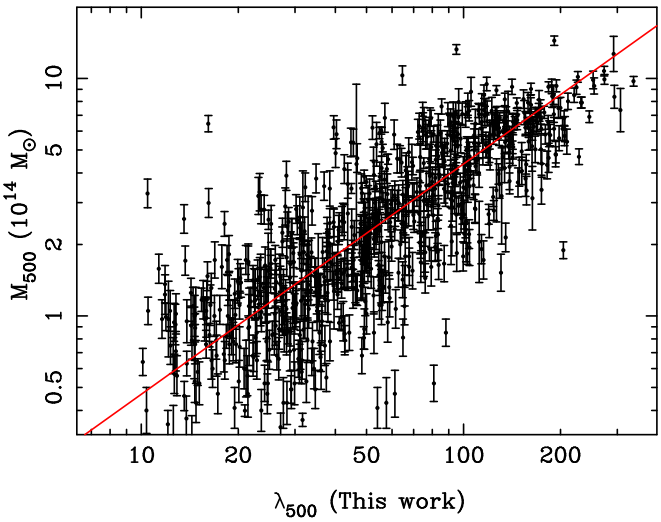


Figure 9. Correlation between cluster richness and the cluster masses compiled in Wen & Han (2015). The solid line is the best fit to the data.

clusters within a redshift difference of $0.05(1+z)$, giving a detection rate of at least 99%. A careful check shows that another 35 Planck clusters have a poor redshift estimate and can be matched within a larger redshift difference (Wen & Han 2022). One undetected cluster has a total number of galaxy candidates of $N_{gal} = 5$, below our threshold. We inspect the color images of the other seven clusters and find no BCG-like galaxies and no overdensity of member galaxy candidates in the given redshift slice. They are probably false detections of the SZ clusters.

The ACT has detected galaxy clusters at the frequencies of 148 GHz, 218 GHz, and 227 GHz, with an angular resolution of

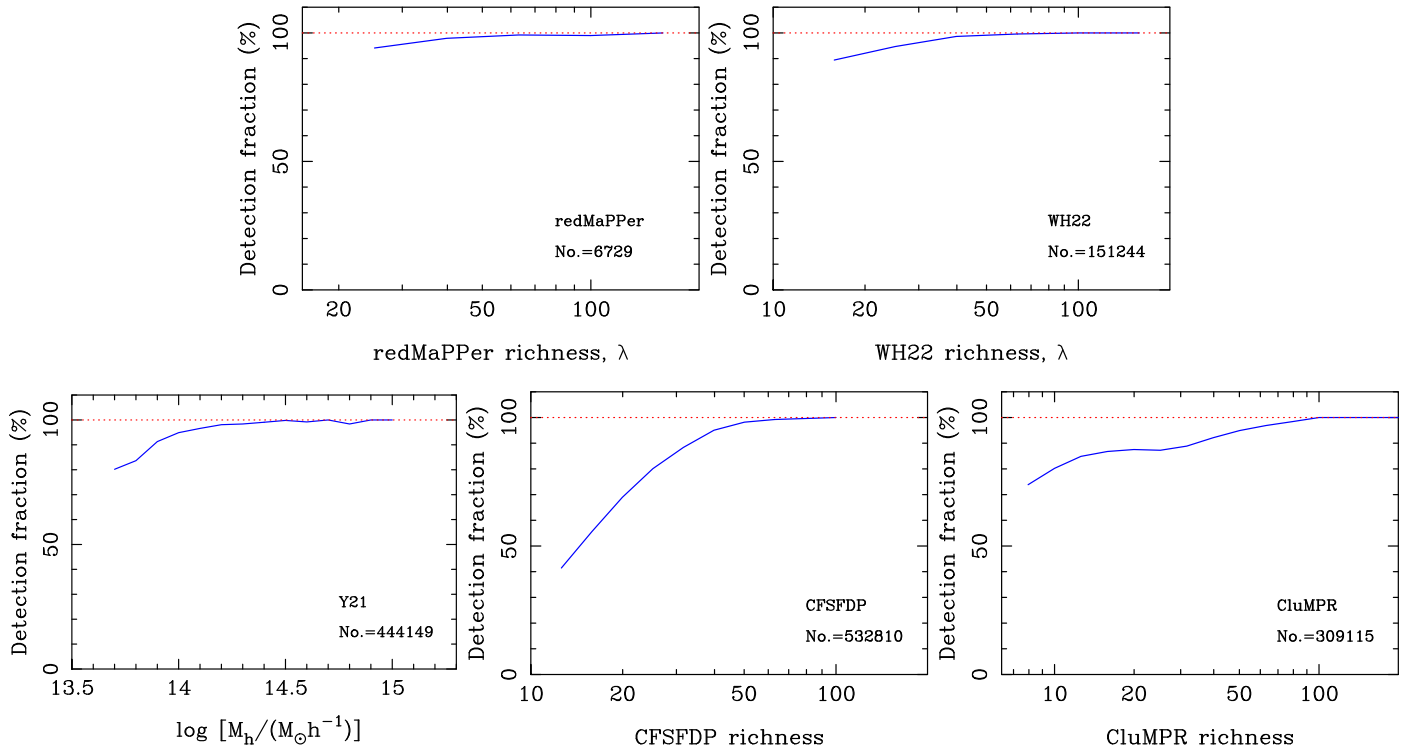


Figure 10. The detection fraction of clusters as a function of richness or halo mass for clusters in the redMaPPer (Rykoff et al. 2016), WH22 (Wen & Han 2022), Y21 (Yang et al. 2021), CFSFDP (Zou et al. 2022) and CluMPR (Yantovski-Barth et al. 2024) catalogs.

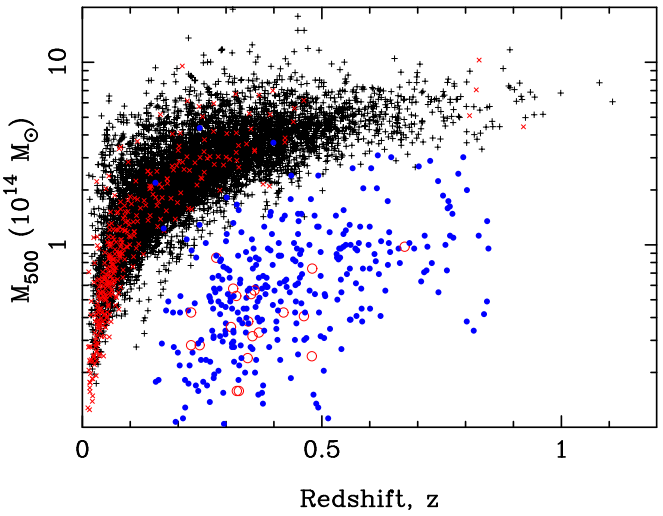


Figure 11. Mass–redshift diagram for clusters in the RASS-MCMF (Klein et al. 2023) and eROSITA catalogs (Bulbul et al. 2022). The matched RASS-MCMF clusters are shown by black plus symbols and the unmatched ones by red crosses. The matched eROSITA clusters are shown by blue dots and the unmatched ones by red circles.

1.′4 at 148 GHz (Swetz et al. 2011). The latest ACT SZ catalog includes 4195 massive clusters of $M_{500} > 1.5 \times 10^{14} M_{\odot}$ at redshifts $0.04 < z < 1.95$ from a sky area of 13,211 deg^2 (Hilton et al. 2021), of which 3914 clusters are located in the DESI Legacy Surveys footprint. The SPT is another excellent facility for detecting SZ clusters at the frequencies of 95 GHz, 150 GHz, and 220 GHz (Bleem et al. 2015). The SPT cluster catalog contains 1442 massive clusters up to redshift ∼1.6 from three surveys with different regions and depths (Bleem et al. 2015, 2020; Huang et al. 2020; Bleem et al. 2023). We detect 95% of ACT clusters and 93% of SPT clusters in our optical cluster sample. The other nondetected clusters are mostly at high redshifts of $z > 1$.

5. Evolution of Dynamical States of Clusters and BCGs

Following Wen & Han (2013), we assess the dynamical states of rich clusters by using the cluster substructures shown by member galaxy distribution. In addition, we discuss the growth of BCG stellar masses with redshift.

5.1. Dynamical States of Clusters

Dynamical states are fundamental for many studies of galaxy clusters because they contain information on the assembly history of clusters (Smith & Taylor 2008). The dynamical parameters affect the mass determination of clusters in cosmological studies (Smith et al. 2003) and constrain astrophysics processes inside clusters, e.g., gas cooling, heating, and electron acceleration (Burns et al. 2008; Feretti et al. 2012). Clusters have various dynamical states and are often classified simply as relaxed clusters and unrelaxed clusters. About 30%–80% of clusters show significant substructures in optical or X-ray, and they are in unrelaxed states (e.g., Dressler & Shectman 1988; Santos et al. 2008; De Luca et al. 2021). It has been suggested that dynamical states and substructures significantly evolve with redshift (e.g., Maughan et al. 2008; Mann & Ebeling 2012), but this is not conclusive (e.g., Weißmann et al. 2013; Wen & Han 2013; Ghirardini et al. 2022), mainly due to the criteria for relaxed clusters and the bias of the selected samples (Eckert et al. 2011).

Wen & Han (2013) presented a method to measure the dynamical states of galaxy clusters based on optical photometric data. First, a smoothed optical map is obtained from the two-dimensional distribution of member galaxies, weighting

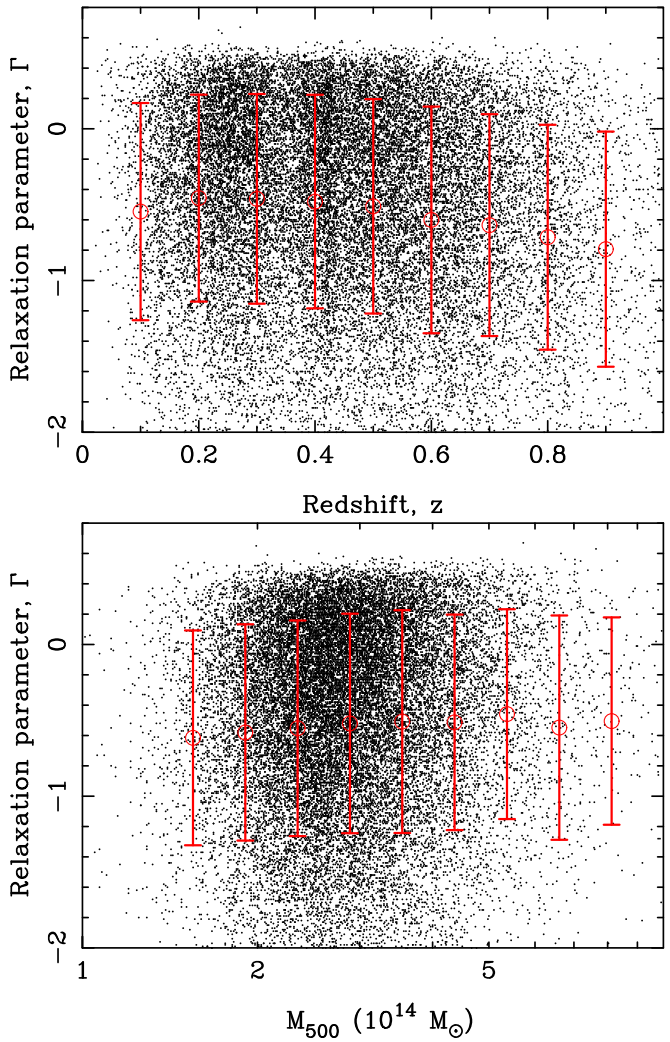


Figure 12. The distribution of the relaxation parameter Γ for 27,685 massive clusters, indicating no significant evolution with redshift (upper panel) and nondependence with cluster mass (lower panel). The open circles and the error bars represent the mean values and scatter of the data set, respectively.

their luminosities. Then, three quantities are calculated: the asymmetry, the ridge flatness, and the normalized deviation of the smoothed map within the region of r_{500}. The asymmetry measures the asymmetry of the distribution of the smoothed optical light around the cluster center. The ridge flatness measures the relative steepness of the most flat light profile in the ridge direction compared to those in the other directions. The normalized deviation measures the deviation of the optical map from a model. After testing a cluster sample, three measurements are combined to define a relaxation parameter, Γ, which has been used to quantify the dynamical states of a sample of 2092 rich clusters. This relaxation parameter is well correlated with the dynamical parameters obtained from X-ray measurements (Wen & Han 2013; Yuan & Han 2020).

For 27,685 rich clusters in our catalog that have more than 30 member galaxy candidates, we calculate their Γ values as listed in Table 2. This is the largest sample of optical clusters with dynamical states quantified. The distribution of Γ values is shown in Figure 12. The dynamical states of clusters have a wide distribution of Γ values, not a bimodal distribution for relaxed and unrelaxed states. No significant variation of Γ with redshift or cluster mass is found, consistent with the conclusions we previously obtained from the SDSS clusters (Wen & Han 2013) and Nurgaliev et al. (2017) obtained from an X-ray cluster sample.

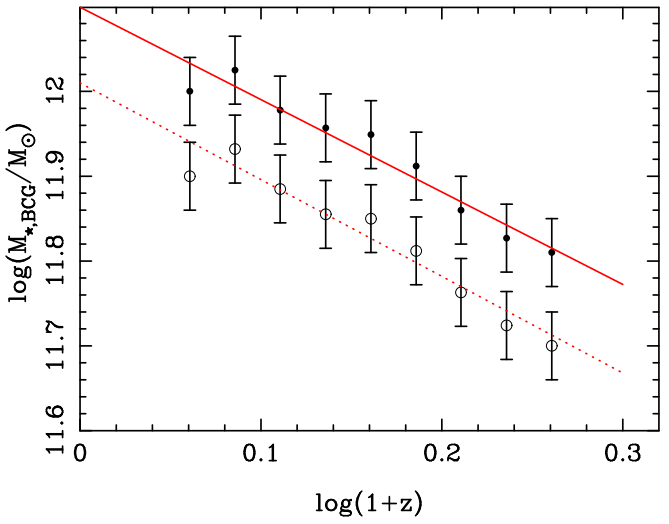


Figure 13. To keep a constant comoving number density of BCGs, the stellar mass threshold of BCGs is shown as a function of redshift. The dots and open circles represent the thresholds, starting from reference values of $M_{\star,\mathrm{BCG}} = 10^{12}\,M_\odot$ and $10^{11.9}\,M_\odot$ at $z = 0.1$–0.2, respectively. The solid and dashed lines are the best fits to the data shown by the dots and open circles, respectively.

If the relaxed and unrelaxed clusters have to be divided literally by separating them at $\Gamma = 0$, one can get 26.5% of the rich clusters identified in this paper being dynamically relaxed, which is consistent with 28% for the rich SDSS clusters from Wen & Han (2013).

5.2. Growth of BCGs

The BCGs have intriguing properties, and their evolution is tightly related to the host clusters (Von Der Linden et al. 2007; Wen et al. 2012; Yuan & Wen 2022). The stellar mass of BCGs can be a good tracer for understanding their evolution. From simulations by De Lucia & Blaizot (2007), the BCG stellar mass grows by a factor of 3 from $z = 1$ to $z = 0$. Diverse results were later obtained by observations (e.g., Lin et al. 2013; Zhang et al. 2016; Wen & Han 2021). The infrared luminosities were found to be consistent with a passive evolution model since $z \sim 1$, suggesting no significant change in the BCG stellar masses (Whiley et al. 2008; Stott et al. 2010). Lidman et al. (2012) corrected the relation between BCG stellar mass and cluster mass, and found that the stellar mass of BCGs increases by a factor of 1.8 ± 0.3 from $z = 0.9$ to 0.2.

Here, we investigate the growth of the BCG stellar mass by a new approach. Assuming that mergers between BCGs are negligible, the comoving number density of BCGs should be a constant over cosmic time for a volume-limited complete sample. Since all BCGs in our large cluster sample have a mass $M_{\star,\mathrm{BCG}} > 10^{11}\,M_\odot$, we can take a high threshold of BCG stellar mass for clusters in the redshift range of $z = 0.1$–0.2 as the reference and then reduce the threshold of BCG stellar mass to keep the same comoving density of BCGs in other redshift bins, as shown by Figure 13. For BCGs with a threshold of BCG stellar mass $M_{\star,\mathrm{BCG}} > 10^{12}\,M_\odot$ in the redshift range of $z = 0.1$–0.2, the thresholds in the other redshift bins can be

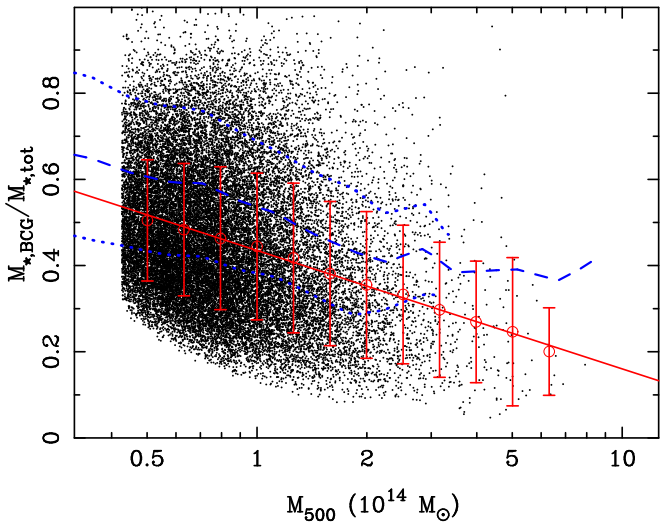


Figure 14. The fraction of BCG stellar mass among the total stellar mass of galaxies in clusters, plotted as a function of cluster mass. The average and data dispersions are given by the circles and error bars, with a fitted line. The dashed line and dotted lines come from the simulations by Kravtsov et al. (2018) for the mean fraction and the dispersion.

fitted by a linear law in the log-log space:

$$\log(M_{\star,\mathrm{BCG}}) = (12.10 \pm 0.02) - (1.09 \pm 0.10)\log(1+z). \quad (5)$$

This relation implies that the stellar mass grows by a factor of $2.12^{+0.16}_{-0.14}$ since $z = 1$. If a threshold of $M_{\star,\mathrm{BCG}} = 10^{11.9}\,M_\odot$ is chosen, one can get a fit to the thresholds with

$$\log(M_{\star,\mathrm{BCG}}) = (12.01 \pm 0.02) - (1.14 \pm 0.11)\log(1+z), \quad (6)$$

which means that the stellar masses of BCGs grow by a factor of $2.20^{+0.17}_{-0.16}$ since $z = 1$. Our results are consistent with previous works based on observation data (Lidman et al. 2012; Zhang et al. 2016; Wen & Han 2021).

In the optical, the BCG is usually dominant inside a cluster. The dominance of BCGs in clusters can be expressed by the fraction of BCG stellar mass relative to the total stellar mass (including the BCG) if the members of the clusters are highly complete. We therefore take all clusters at $z < 0.5$ for this work. As seen in Figure 14, the fraction of BCG stellar mass decreases with cluster mass, indicating that the BCGs are less dominant in more massive clusters with more member galaxies, consistent with the results from simulations (Kravtsov et al. 2018). The mean fraction depends on cluster mass by the relation

$$\frac{M_{\star,\mathrm{BCG}}}{M_{\star,\mathrm{tot}}} = (0.435 \pm 0.003) - (0.275 \pm 0.007)\log M_{500}. \quad (7)$$

In addition, we investigate how many BCG-like galaxies are located in the clusters. BCG-like galaxies are considered as being cluster members if they have a projected separation within $1.5\,r_{500}$ and a redshift difference less than Δ_z. We find that about 76% of massive galaxies with a mass of $M_\star \sim 10^{12}\,M_\odot$, or about 50% of galaxies of $M_\star \sim 10^{11.7}\,M_\odot$ or 10% at $M_\star \sim 10^{11}\,M_\odot$, are located in the identified clusters. Therefore, finding out massive BCG-like galaxies is a useful first step to identifying galaxy clusters.

6. Summary

By using photometric redshifts and the available spectroscopic redshifts of galaxies in the joint DESI Legacy Surveys and WISE data, we obtain a large catalog of 1.58 million galaxy clusters and investigate their properties.

We first estimate the photometric redshifts for 295.6 million galaxies in the latest released data in the *griz*W1W2 bands by using the nearest-neighbor algorithm. The redshift uncertainty is about 0.013 for bright galaxies of $z_\mathrm{m} < 21$ and increases to 0.053 for faint galaxies of $z_\mathrm{m} > 22$. The stellar mass is estimated from the *z*W1-band luminosities for the galaxies in the DR10 and also those galaxies in the DR9 if their photometric redshifts are estimated.

Then, we identify a large sample of 1.58 million galaxy clusters based on the overdensity of total galaxy stellar mass within a redshift slice around BCG-like galaxies. Among them, 877,806 clusters are identified for the first time. The redshift distribution extends to $z < 1.5$, and 338,841 clusters have spectroscopic redshifts. The uncertainty of the cluster photometric redshift is about 0.010. The cluster richness has a good correlation with cluster mass, which indicates that the identified clusters have an equivalent mass of $M_{500} \geqslant 0.47 \times 10^{14}\,M_\odot$. Our cluster catalog includes most of the massive clusters in previous optical, X-ray, and SZ cluster catalogs.

We assess the dynamical states for 27,685 rich clusters based on the substructures of the distribution of cluster member galaxies. No significant evolution of dynamical states is shown. In addition, we find that the stellar mass of BCGs increases by a factor of 2 after $z = 1$. The fraction for the BCG stellar mass relative to the total stellar mass decreases with cluster mass, implying less dominance of BCGs in richer clusters.

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